pfi

Python for Fixed Income

COURSE 5C · RISK MODELS

Correlated Defaults and the Loss Distribution

WEEK 1 · DAY 3

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Today: defaults that move together

The cohort

4,000 invented firms: 18,503 firm-years, 345 defaults

The book

200 invented bonds, each given its notch's fitted rate

Two inputs

Asset correlation 0.20; LGD a beta draw with mean 0.6

Today's question

Once defaults move together, how bad can a year get for the book?

Day 1's independent defaults made a bad year very rare. Real defaults cluster.

Development study years 1 to 14 only. The held-out years open in the capstone.

Names for a default model

Default rates describe invented firms, counted in study years, not calendar years. No agency default study appears.

Allowed names

fitted hazard

a synthetic firm's default chance

survival in study years

Never called

a rating view

a real issuer's default chance

a forecast of defaults

The correlation and the loss given default are inputs chosen for the lesson.

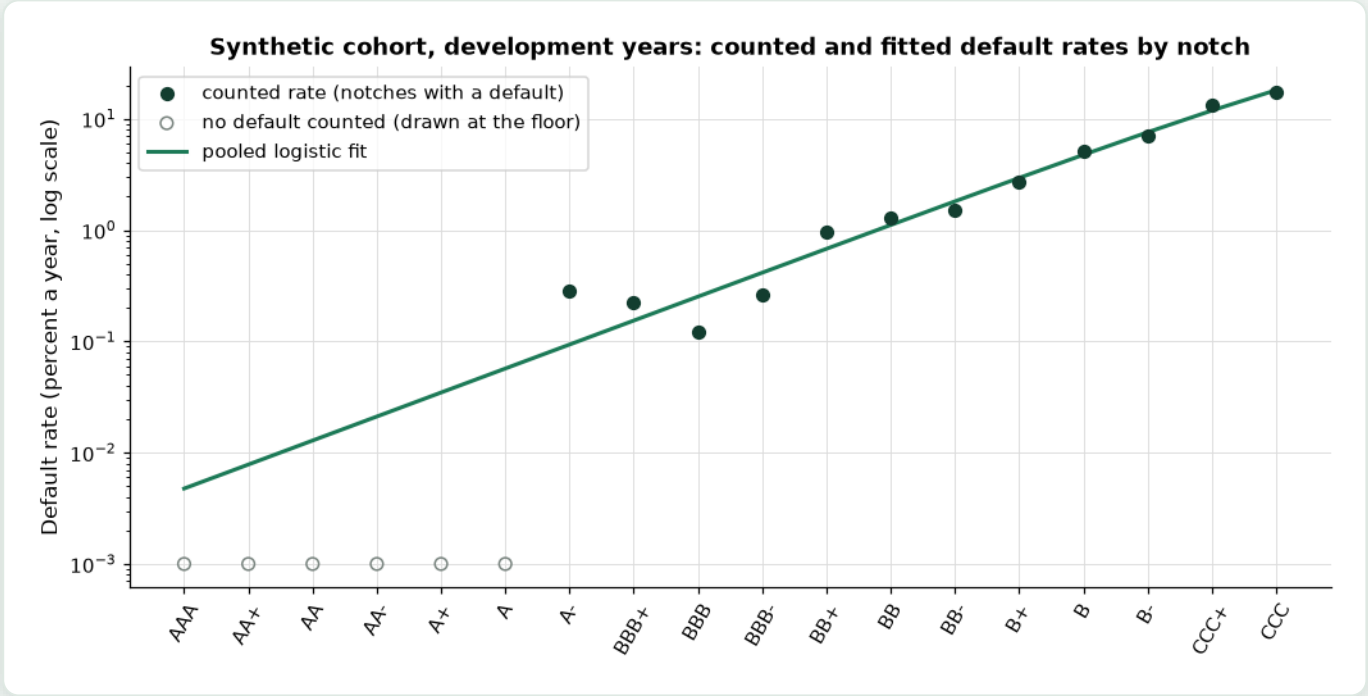
How a default rate is counted

The problem	A default probability for each rating, from data, not assumed
Firm-years	One firm watched for one year; a firm watched 6 years counts 6
The rate	Defaults in a group over its firm-years at risk
You control	Which years are counted, and how firms are grouped
What comes out	A default rate a year per group, with the firm-years behind it

Default rates and the low-default problem

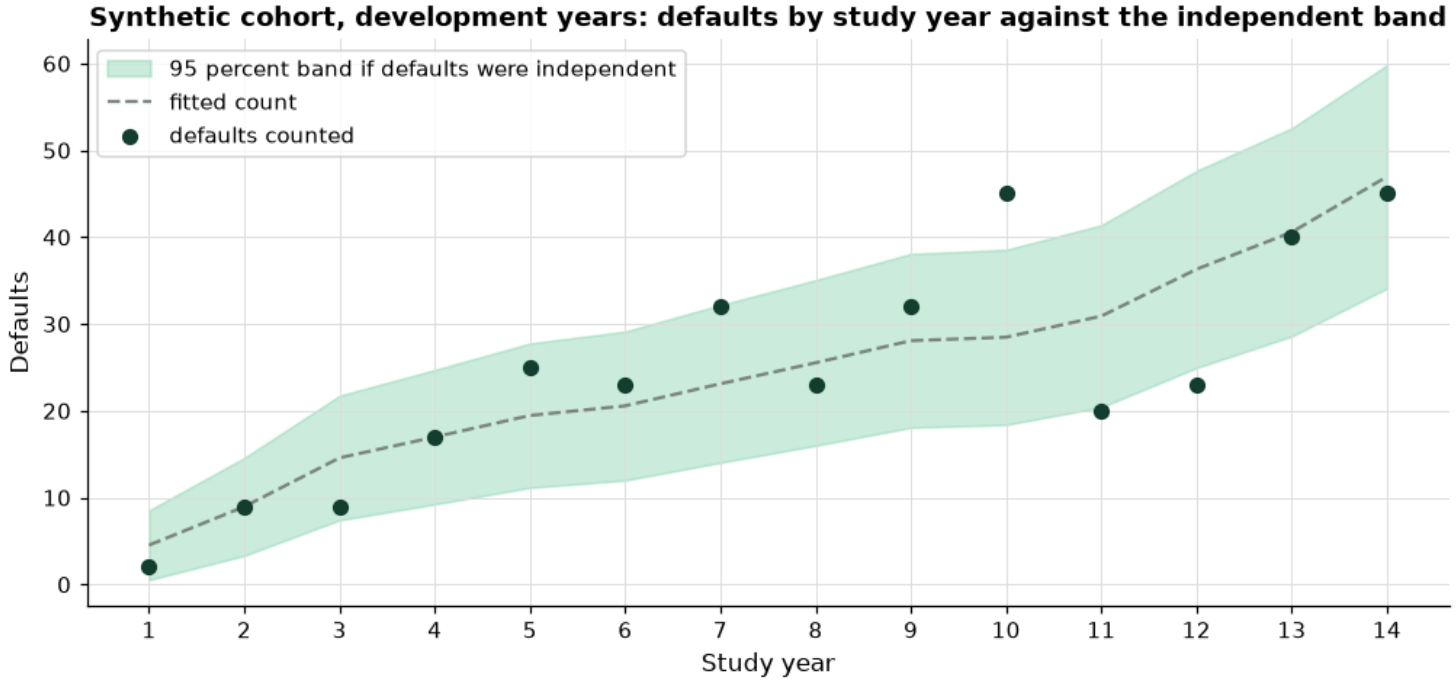
Bucket	Firm-years	Defaults	% a year
AAA	168	0	0.00
AA	1,274	0	0.00
A	3,014	3	0.10
BBB	4,897	10	0.20
BB	5,142	64	1.25
B	3,207	149	4.65
CCC	801	119	14.86

A pooled logistic fit: each notch multiplies the odds by e^0.4967. AA's fitted rate: 0.0129 percent a year.



A count of zero is not a chance of zero. The fitted line is the only word on the top notches.

Defaults come in clusters



3 of 14

years outside

2.17x

independent variance

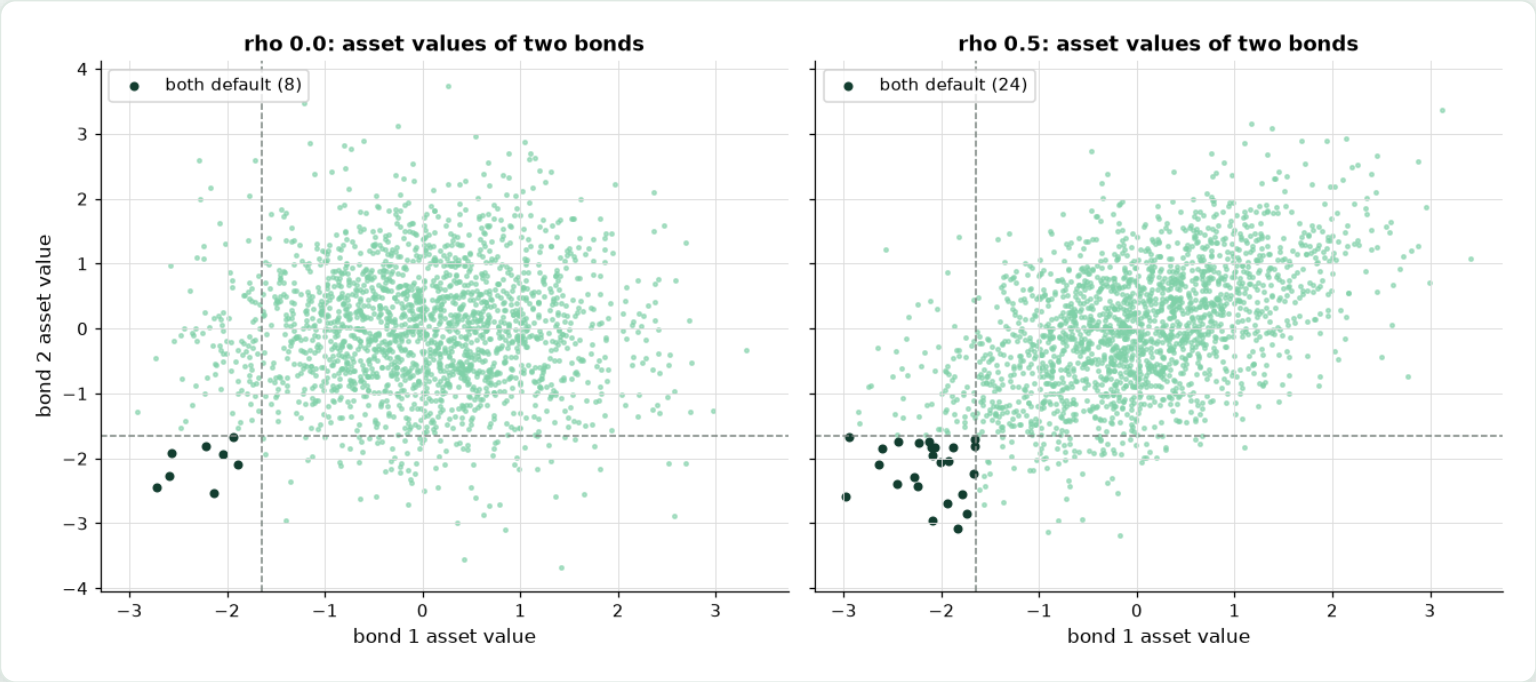
About 1 if firms defaulted independently. The cohort was made with a common factor in each study year, shared by every firm.

The one-factor model puts that clustering into a simulation.

How the one-factor model works

The problem	Defaults that move together, with each bond keeping its own default chance
Asset value	sqrt(rho) x a common factor Z + sqrt(1 - rho) x the bond's own factor
Default	Below the threshold norm.ppf(pd): each bond still defaults with its own pd
You control	rho, the asset correlation: 0 is independent; larger is more bad years
What comes out	A scenarios by bonds table of defaults (one_factor_defaults)

Two bonds, two correlations



0.40%

both, rho 0

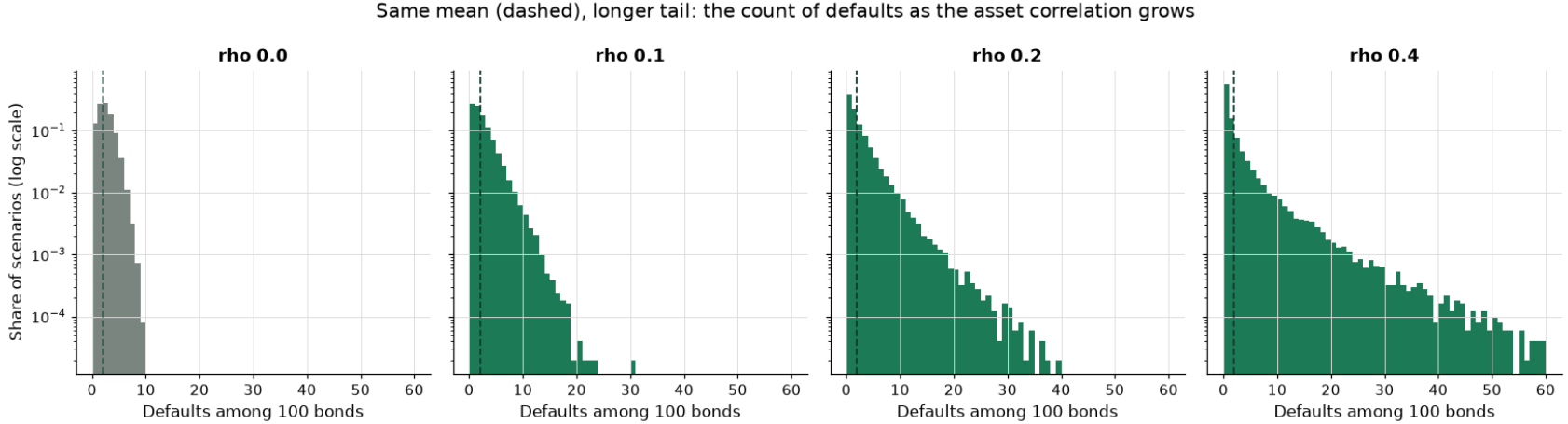
1.20%

both, rho 0.5

Each bond defaults about 5 percent of the time at both correlations; 0.25 percent if independent.

The bonds' own rates have not moved. They now fail together.

Same mean, longer tail



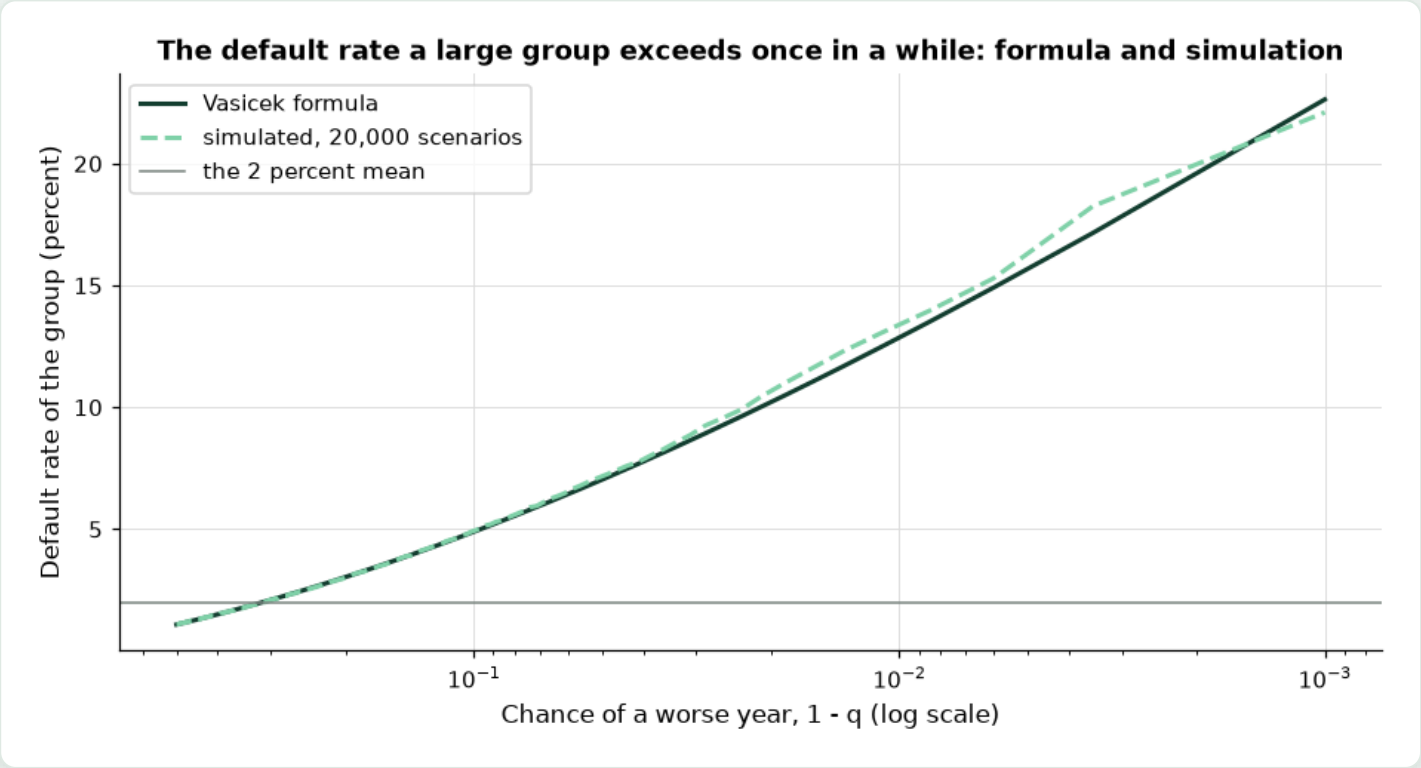
rho	Mean	Std dev	99th pct	10 or more	None
0	2.01	1.40	6	0.00%	13.2%
0.1	2.00	2.19	10	1.17%	27.0%
0.2	2.01	3.01	14	3.10%	38.4%
0.4	2.00	4.63	23	5.43%	56.4%

Correlation does not add defaults. It moves them into the same years: more years with many, and more with none.

How the Vasicek formula works

The problem	The tail of a large group's default rate without a simulation
Large and alike	Each bond's own factor averages out; the common factor alone sets the share
The quantile	N((N^-1(pd) + sqrt(rho) N^-1(q)) / sqrt(1 - rho))
You control	pd, rho, and q
What comes out	The rate exceeded with chance 1 - q (vasicek_quantile); exact only in the limit

The formula against a simulation



q	Simulated %	Vasicek %
0.5	1.10	1.08
0.9	4.95	4.89
0.99	13.40	12.86
0.999	22.10	22.63

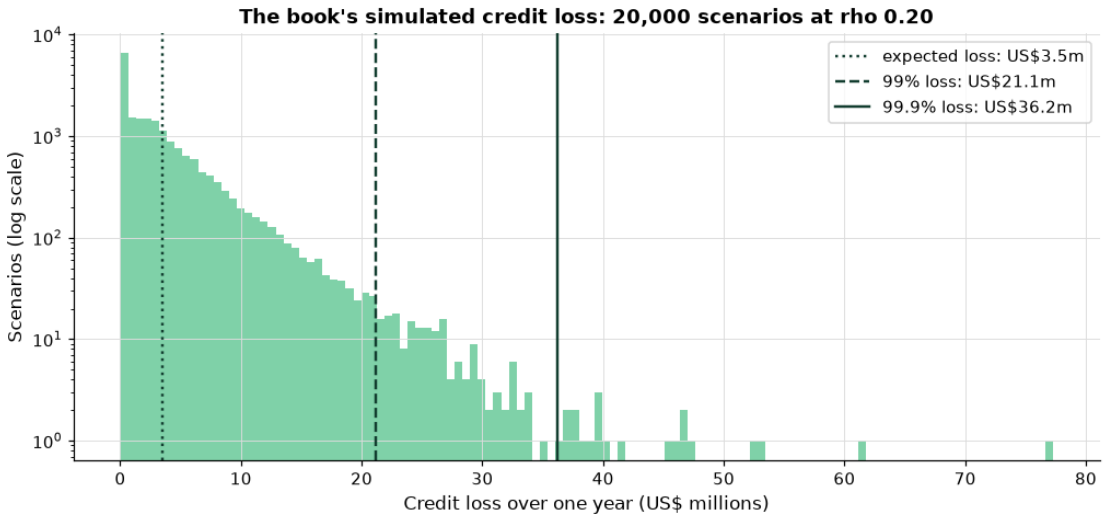
2,000 bonds at 2 percent, rho 0.2. The 99.9 percent rate is the 20th worst of 20,000 scenarios.

A once-in-a-thousand-years rate is more than ten times the average one.

How a loss distribution works

The problem	The book's credit loss over a year, with defaults that cluster
Defaults	The one-factor model with each bond's fitted default chance
Loss	A defaulted bond loses market value x an LGD drawn from a beta (loss_distribution)
You control	rho, the LGD distribution, and the number of scenarios
What comes out	One loss per scenario: the mean is the expected loss, the tail is credit VaR

The book's credit loss



US$3.5m

mean: exact US$3.5m

US$36.2m

99.9% loss

US$45.2m

beyond it, on average

29.2%

with no default

Under these inputs. A credit loss distribution is a heap near zero and a long tail to the right.

Where the tail sits, and what rho does

Sector	Worst 1%	Share
Materials	US$5.3m	19.4%
Industrials	US$4.4m	15.9%
Financials	US$3.3m	12.0%
Consumer	US$3.2m	11.8%
Telecom	US$2.8m	10.3%

The five sectors with the largest share of the worst 1 percent of scenarios: high yield bonds sit closest to their thresholds.

rho	Mean loss	99.9% loss
0	US$3.5m	US$14.4m
0.1	US$3.5m	US$24.5m
0.2	US$3.5m	US$35.7m
0.3	US$3.5m	US$49.8m

The mean barely moves; the 99.9 percent loss grows several times over.

The correlation is the setting the tail depends on most. Sweep it first.

Correlated factors and Cholesky

```
L = np.linalg.cholesky(corr_two)
draws = rng.standard_normal((n, 2)) @ L.T
# L @ L.T == corr_two
```

0.4957

drawn correlation, 0.5 asked for

Three correlations typed one pair at a time

A and B at 0.9, A and C at -0.6, B and C at 0.4: no three variables can do that.

Smallest eigenvalue -0.2903

Cholesky refused the matrix: Matrix is not positive definite

Excel has no Cholesky function. The error is the check doing its job.

The formula in the Basel rules

Basel Committee on Banking Supervision, July 2005

Explanatory Note on the IRB Risk Weight Functions

Confidence level 99.9%; corporate correlations between 12% and 24%

"With a view on Merton's and Vasicek's ground work, the Basel Committee decided to adopt the assumptions of a normal distribution for the systematic and idiosyncratic risk factors."

Source: BCBS, An Explanatory Note on the Basel II IRB Risk Weight Functions, July 2005, pages 5, 6, 12, and 13 (bis.org). It cites Vasicek, Loan portfolio value, Risk, December 2002.

Excel side by side

Excel side by side (tested in Excel)

`=AVERAGEIFS(B2:B18504,A2:A18504,"BB")`	BB rate 1.245%
`=1/(1+EXP(-($J$1+$J$2*12)))`	fitted BB 1.112%
`=SUMPRODUCT(D2:D201,E2:E201)*0.6`	expected loss US$3,482,412
`=1-BINOM.DIST(9,100,0.02,TRUE)`	0.000034, independent
`=NORM.S.DIST((NORM.S.INV(0.02)+SQRT(0.2)*NORM.S.INV(0.999))/SQRT(1-0.2),TRUE)`	22.63%
`=BETA.INV(G2,6,4)`	an LGD draw: beta.ppf within 1e-9

Excel has no logistic regression and no Cholesky; given the coefficients, the fitted rate is one EXP.

Save the day: look before you commit

1. Stage

git add the four code files

2. Look

git diff --cached --stat lists what the commit will hold

3. Commit

Then commit, and read the log

```
git diff --cached --stat
```

The cohort, the book, and plotkit.py stay out of the staged list.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

What is drawn, from which inputs, in which unit: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: a "common" factor drawn once per bond: the count's variance is 1.96, the binomial one.

Code that runs is not the same as code that is right.

What you can do now

- Count default rates from firm-years; fill the low-default buckets
- Show that defaults cluster by year
- Simulate defaults through one common factor (one_factor_defaults)
- Check against the Vasicek formula (vasicek_quantile)
- Simulate the book's credit loss (loss_distribution); 18 tests pass

Week 1, Day 4 Survival curves and Kaplan-Meier